1. **Random and Systematic Error**

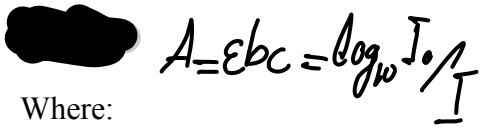
Random error	Systematic error
Random error is unpredictable. It can't be control It is caused by minor difference in the sample. Data is scattered more or less symmetrically	Systematic error is consistent and repeatable. It can be minimized Caused by inaccuracy in the system. All the measurement From true measurement.

2. Write a note on UV spectroscopy

UV Spectroscopy

Introduction
Ultraviolet (UV) spectroscopy is an analytical technique used to study the absorption of ultraviolet light by chemical substances. This method is based on the interaction of UV radiation (typically 200-400 nm) with molecules, causing electronic transitions. It is widely used in chemical, pharmaceutical, and biological analysis to determine molecular structure, concentration, and functional groups.

Principle
UV spectroscopy operates on the **Beer-Lambert Law**, which states:



Where:

- **A** = Absorbance
- **ε** = Molar absorptivity (L/mock)
- **b** = Path length (cm)
- **C** = Concentration (moll/L)

A molecule absorbs UV light when the energy of the incident radiation matches the energy difference between its electronic ground state and an excited state.

Electronic Transitions in UV Spectroscopy
Different types of electronic transitions occur depending on the molecular structure:

- **σ → σ*** (High energy, below 200 nm)

- **n → σ*** (Present in molecules with lone pairs, e.g., O, N, S)
- **π → π*** (Observed in conjugated double bonds)
- **n → π*** (Characteristic of carbonyl compounds)

Instrumentation
a

UV-Vis spectrophotometer consists of:

1. **Light Source** (Deuterium lamp for UV and Tungsten lamp for visible region)
2.
3. (Prism or diffraction grating to isolate specific wavelengths)
4. **Sample Holder** (Cuvette made of quartz or glass)
5. **Detector** (Photodiode or photomultiplier tube)
6. **Readout System** (Digital display or computer interface)

Applications of UV Spectroscopy

1. **Structure Elucidation**: Helps identify functional groups and
2. .
3. **Quantitative Analysis**: Used in drug testing, food analysis, and chemical concentration measurement.
4. **Kinetics Studies**: Monitoring reaction rates and mechanisms.
5. **Biochemical Analysis**: Used for DNA, RNA, and protein quantification.
6. **Pharmaceutical Quality Control**: Checking purity, stability, and degradation of drugs.

3. **Aldehyde and Ketones**

The chemical entities having C=O functional group in their structure are known as carbonyl compounds. In case of aldehydes, carbonyl group occurs at the end of the carbon chain whereas in ketones, it lies in the middle. Due to this, the carbon atom in the carbonyl group has also formed bonds with hydrogen and another carbon in case of aldehydes and with two other carbons in case of ketones.

Ketones and aldehydes mostly display a prominent and strong stake shaped band between 1710 to 1720 cm^{-1} due to the presence of this immensely polar C=O bond. This band usually appears in the middle of the IR spectrum and due to its size, position and shape; it's hard to be ignored. Aldehydes also possess C-H bond made with the sp2 hybridized carbon of C=O bond. As a result, they also show a couple of medium strength bands lying between 2700 & 2800cm^{-1}. Ketones do not contain these bands in their IR spectrum as their sp2 carbon atom lacks such C-H bonds. This difference enables us to distinguish between aldehydes and ketones.

4. **Note on ICP/OES**

ICP-OES or inductively coupled plasma optical emission spectroscopy is a detection technique which produces high energy atoms (excited atoms) & ions with the help of inductively coupled plasma. These excited atoms or ions then emit photons of a specific wavelength while coming back to their ground states. The emitted wavelength is characteristic of the element from which the atoms or ions are produced. Plasma serves as a high temperature source of ionized argon or any other source gas. Cooled electrical coils play an important role to sustain and maintain plasma by inductive coupling. The plasma can attain any temperature within the range 6000-10000 K. The concentration of the target element in the sample tends to be directly proportional.

Working: The **ICP torch** consists of multiple quartz glass tubes aligned side by side, with a portion surrounded by an RF coil powered by a **radio-frequency generator**. When the torch is activated, a high-energy RF signal creates an intense electromagnetic field, while a **Tesla unit** ignites the argon gas flowing through the torch, initiating ionization. Once the plasma is formed, the Tesla unit is turned off.

The ionized argon gas flows rotationally within the RF coil's magnetic field, maintaining stable **plasma at ~7000K** through continuous collisions between charged and neutral argon particles. A **peristaltic pump** delivers the organic or aqueous sample to an analytical nebulizer, converting it into a fine mist before introducing it into the plasma flame. The intense energy of the plasma breaks down the sample into **charged ions and free atoms**, which undergo repeated ionization and recombination, emitting **characteristic radiation wavelengths** for elemental analysis.

5. **Which factor retards the biomolecules in the stationary phase? **

- Sieve **mechanism** (size-based separation)
- Charge **interaction** (electrostatic attraction/repulsion)
- Solubility **characteristics** (polarity-dependent retention)
- Biological **specificity** (affinity chromatography

6. **Hollow Cathode Lamp**

Hollow cathode lamp is also known as HCL, is a type of a vacuum discharged lamp that severs as a source of the light in atomic absorption spectroscopy. It generates photons of particular metal for which cathode is made. It is composed of cathode and some noble gas at very low pressure. HCL has advantage over the cold cathode as it permits conduction at lower voltage.

7. **Sublimation**

• **Definition**: Transition of a substance from solid to gas without passing through the liquid phase.
• **Example**: Separation of benzoic acid or naphthalene from salt.

8. **In acid-base titration, what is the titrant? **

• The **titrant** is the known solution added to the analyte.
• Typically a strong acid (HCl) or strong base (NaOH) is used

9. **Lambert Equation**

• The Beer-Lambert Law equation: A=εbC
• It relates absorbance to concentration and path length

Q5: Difference between filtration and sublimation?

Filtration	Sublimation
Separation of solid insoluble particles from solution.	Separation of one solid from other if it vaporizes by heating.
For example separation of sand from water	For example separation of salt from benzoic acid or naphthalene.
Uses a porous medium (e.g., filter paper) for separation.	No medium Is required; it occurs under specific conditions.

12. **Complementary colors**

Complementary Colors

Complementary colors are pairs of colors which, when combined or mixed, cancel each other out by producing a gray scale color like white or black. When placed next to each other, they create the strongest contrast for those two colors the three traditional sets of complementary colors, as derived from the Red-Yellow-Blue color model, are red and green, yellow and purple, and orange and blue

13. **Scatter diagram**

Scatter diagram, also referred to as a scatter plot, is a quality control tool used to analyze the relationships between two different variables and to find out how much the two of them affect each other. One variable is taken on the vertical axis while the other is plotted on the horizontal axis. Their intersecting points can give a graphical insight on the relationship patterns. Scatter plot can be frequently used for:

- Validating two different variables.
- Analyzing trends in relation to time
- Confirming a hypothesis about two different variables

14. solid phase extraction:

Solid-phase extraction is an extractive technique by which compounds that are dissolved or suspended in a liquid mixture are separated from other compounds in the mixture according to their physical and chemical properties. Solid phase extraction (SPE) is a technique designed for rapid, selective sample preparation and purification prior to the chromatographic analysis (e.g. HPLC, GC, TLC). In SPE, one or more analytes from a liquid sample are isolated by extracting, partitioning, and/or adsorbing onto a solid stationary phase.

17. **Why do transition metals show color in organic compounds? **

Transition metals are coloured because of d-d transitions while organic molecules are coloured because of conjugation.

21. **Mechanism of IR spectroscopy?*

Infrared spectroscopy or vibrational spectroscopy is an analytical technique used to identify and study functional groups or chemical compounds by measuring the degree of interaction of infrared radiations with them. An instrument called IR spectrometer generates an infrared spectrum upon the interaction of IR radiations with a known or unknown sample in its gaseous, liquid or solid state. IR spectrum is displayed in the form of a graph with wavelength or frequency along horizontal axis and absorbance or transmittance along vertical axis. The IR region usually exists between 0.7 to 400μm wavelength range of electromagnetic spectrum.

PRINCIPLE OF IR SPECTROSCOPY

The principle underlying IR Spectroscopy is that molecules tend to absorb frequencies based on their unique structure. Such absorptions usually occur when vibrational frequencies of target molecules match the frequency of absorbed radiations. Molecules have the capacity to vibrate in many forms, also known as their vibrational modes. If a molecule has N atoms in total, then its vibrational modes will be:

- 3N - 5 degrees for linear molecules
- 3N – 6 degrees for nonlinear molecules

In any given sample, a particular vibrational mode will be considered “IR active” if it is associated with alterations in dipole moment. Bands in the IR Spectra will be generated only if the vibrations cause change in polarity

22. **Recovery of acid and base from their salt in an organic compound

By changing organic acids and bases into their salts we can revert the polarity from organic to aqueous soluble, e.g if we have a mixture of benzoic acid and p-methoxyphenol, dissolved in dichloromethane. We can do the either way, add 5% HCl to make amine water soluble or add 5% NaOH to make acid water soluble. We can separate alkaloids from plant extract or organic acids from food in this way.

Polarity is important because it determines whether a molecule is hydrophilic (from the Greek for water-loving) or hydrophobic (from the Greek for water-fearing or water-averse). Molecules with high polarity are hydrophilic, and mix well with other polar compounds such as water

23. **Classical analytical methods**

Classical Analytical methods are the techniques which are the fundamentals of laboratory practices. These are the traditional method of chemical analysis which is still being used by scientists even up to this date. Classical method subdivided into two methods: the qualitative and the quantitative method of analysis

- Acid base titration
- Complexometric titration
- Redox titration
- Gravimetric analysis

24. **Solvent extraction, recovery of solvent from acids and bases**

Solvent extraction

Solvent extraction is the partial removal of a substance from a solution or mixture by dissolving it in another, immiscible solvent in which it is more soluble.

- Acids can be recovered from their salts by treating with HCl
- Decrease pH to acidic range~2
- For salts of bases we have use solution of a base to increase pH~10

25. **Documentation control**

Quality assurance document control is the process used in the management, coordination, control, delivery, or support of an item required for quality assurance purposes.

To Write QA Documentation:

- Make a Test Plan and a Test Progress Report. ...
- Create Test Cases. ...
- Write a Defect Report. ...
- Note these Useful Tips for Defect Report Writing.
- Submit a Defect Report. ...
- To Wrap Up. ...

26. **Gravimetric analysis

. Gravimetric analysis

Gravimetric analysis is a method in analytical chemistry to determine the quantity of analyte based on the mass of a solid. It is the technique of finding the mass of an analyte or chemical compounds that are inter-related chemically. This is the classical method of analysis that provides measurement with reliability and precision. This analysis serves as the most authentic measuring techniques.

Steps followed in the Gravimetric Analysis

- Preparation of a solution containing a known weight of the sample.
- Separation of the desired constituent.
- Weighing the isolated constituent.
- Computation of the amount of the particular constituent in the sample from the observed weight of the isolated substance.

Types of Gravimetric Analysis

There are 4 fundamental types of gravimetric analysis.

- Volatilization Gravimetry involves separating components of our mixture by heating or chemically decomposing the sample.
- Precipitation Gravimetry uses a precipitation reaction to separate one or more parts of a solution by incorporating it into a solid.

- Electrogravimetry is a method used to separate and quantify ions of a substance, usually a metal.
- Thermogravimetric is a method of thermal analysis in which changes in physical and chemical properties of materials are measured as a function of increasing temperature or as a function of time.

29.Properties of good precipitates

- Must be free of contaminants and easily filterable;
- Should produce a precipitate that is sufficiently insoluble that it will not dissolve during washing;
- The precipitate needs to be chemically stable;
- The precipitate needs to be of known composition after drying or even calcination.

30 **Steps in measuring uncertainty

- Identify sources of uncertainty.
- Quantify individual uncertainties.
- Combine uncertainties using statistical methods.
- Report total uncertainty with measurement